Dennis Menn

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Objective

Advancing the **computational efficiency** of **generative AI** with a focus on transformer architecture.

Education

Doctor of Philosophy in Electrical and Computer Engineering — Expected: August 2027
University of Texas at Austin — GPA: 3.75/4.00

Master of Science in Electrical and Computer Engineering — July 2023
National Taiwan University — GPA: 4.26/4.30

Bachelor of Science in Electrical and Computer Engineering — June 2021
National Taiwan University — GPA: 4.03/4.30

Relevant Experiences

Futurewei Research Intern (Austin Office) — May 2025 — August 2025

- Proposed content-aware video editing pipeline that reuses edited duplicated tokens, enabling reduced latency for state-of-the-art streaming video editing models
- This approach can improve latency by up to 3.5× and achieve near-real-time performance (29 fps) on an RTX 6000 GPU, with only mild degradation in editing quality

Research Assistant at Energy-Aware Computing Lab — January 2024 — Present
University of Texas at Austin. Advisor: Diana Marculescu (department chair)

- Designing end-to-end framework for real-time on-device video-to-video generation, enhancing temporal coherence in video editing workflows (ongoing)
- Developed data-efficient algorithms for image anomaly detection, reducing data requirements by 1000x while maintaining classification accuracy to 72.5%

Publications

"Synthesizing Artifact Dataset for Pixel-level Detection", **Dennis Menn**, Feng Liang, Diana Marculescu. WACV 2026 (under review)

"**Similarity Trajectory: Linking Sampling Process to the Quality of Generated Data in Diffusion Models**", **Dennis Menn**, Feng Liang, Hung-Yueh Chiang, Diana Marculescu. WACVW 2025

"**Paint and Relight: A Two-Stage Framework for Refining Neural Radiance Fields**", **Dennis Menn**, Ta-che Hsiao

"**Investigating Human-Identifiable Features Hidden in Adversarial Perturbations**", **Dennis Menn**, Tzu-hsun Feng, Sriram Vishwanath, Hung-yi Lee

"**Can GPT Models Detect Artifacts in Images?**" **Dennis Menn**, Alexandros Dimakis, Diana Marculescu

Projects

Don't Edit Twice: Content-Aware Computational Video Editing Algorithm — July 2025 – Present

- Developed a content-aware video editing pipeline that utilizes spatial and temporal duplication in video data.
- Achieved up to 29 FPS and 3.5x speed up for near real-time video editing using the Causvid model.

High-Level-Synthesis for Implementing Convolution Neural Networks — August 2024 – December 2024

- Architected FPGA-optimized quantized CNN using High-Level Synthesis (HLS), implementing bit-width quantization and developing all convolutional/pooling kernels from scratch
- Achieved 92.1% accuracy on Fashion MNIST despite applying quantization and pruning techniques, optimizing computational resources and reducing model size without sacrificing performance

Image Anomaly Detection for Synthesized Images Through Sampling Process — March 2024 – December 2024

- Established a correlation between denoised image similarity during the sampling process and the quality of generated images
- Reduced the required training data for image artifact detection by 1,000× compared to previous methodologies while maintaining high classification accuracy

Skills

- Programming Languages: Python, C/C++, Matlab, Shell script
- Modules/Tools: Pytorch, SKlearn, OpenCV, SQL, PSpice, HLS, Quartus,